

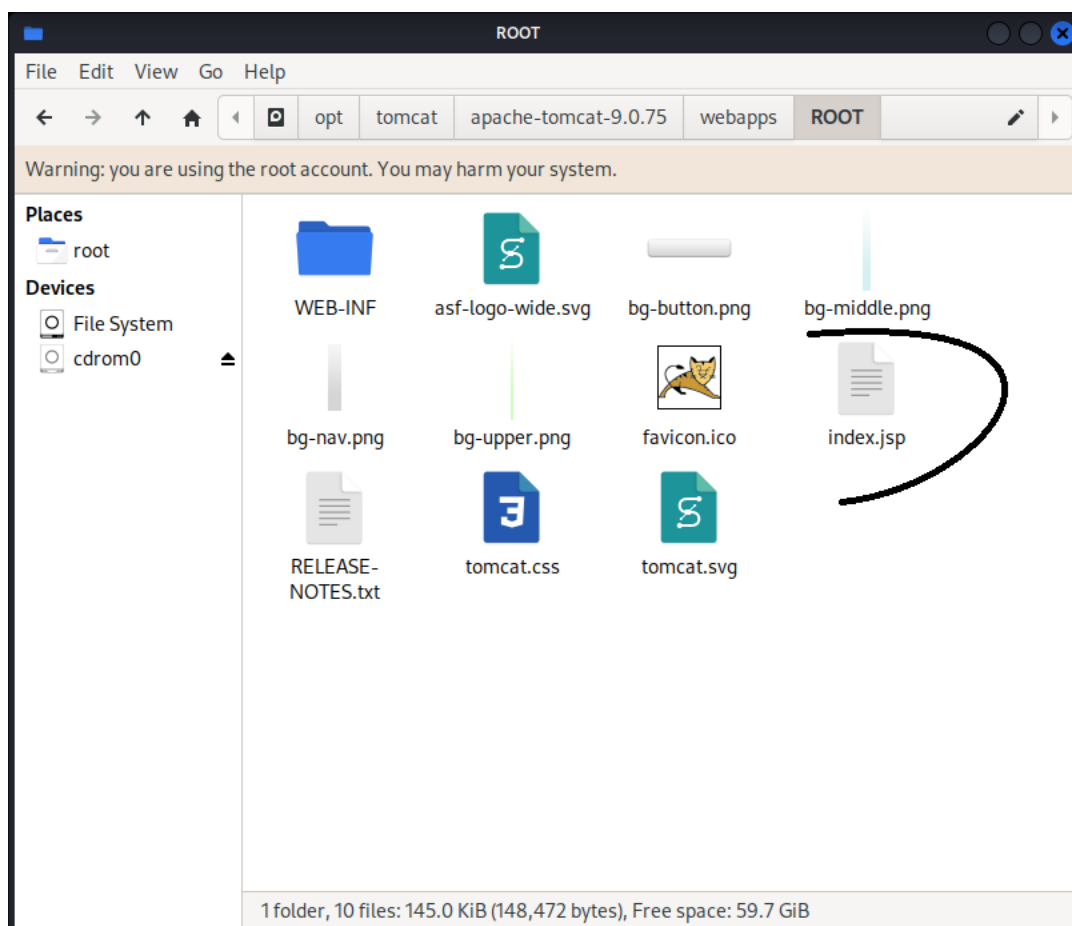
SIT 218/738: Secure coding

Credit task 2.2C: Security configurations of Tomcat server

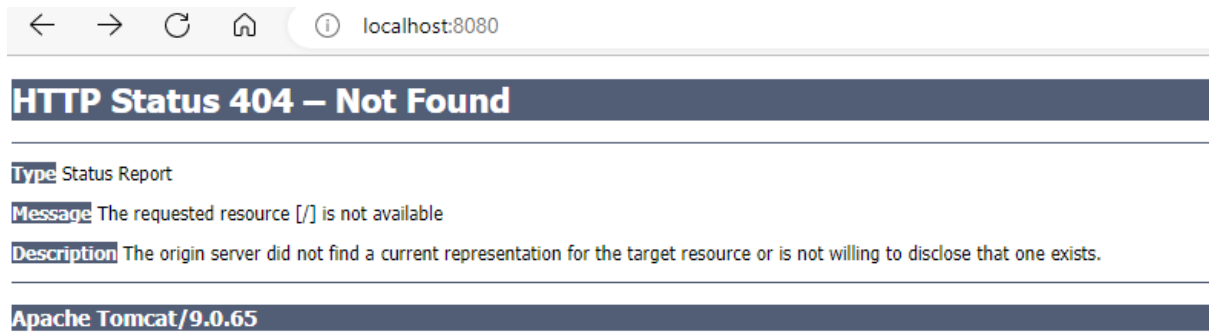
Overview

In the previous task we changed the server banner to send a custom information about our tomcat server. However, you might have noticed the response still consists details of the web server due to the default index page.

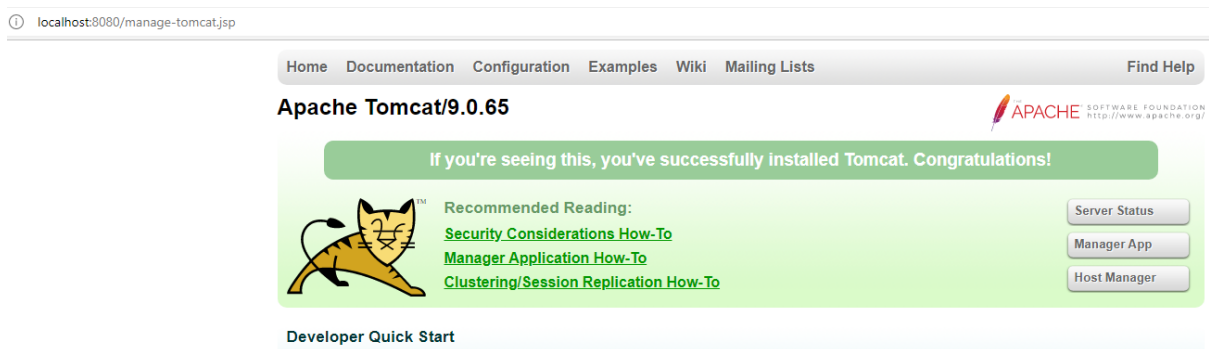
Subtask-1: First you will rename the **index.jsp** in the root to another name “manage-tomcat.jsp”. See screenshot below for location of the index.jsp file.



The default Tomcat manager page should not be accessible from the root path (/ -- eg. Localhost:8080) instead you should get a HTTP status 404 page when accessing the tomcat with the link: <http://localhost:8080> as below.



The admin page should however be accessible from <http://localhost:8080/manage-tomcat.jsp> as per screenshot below.

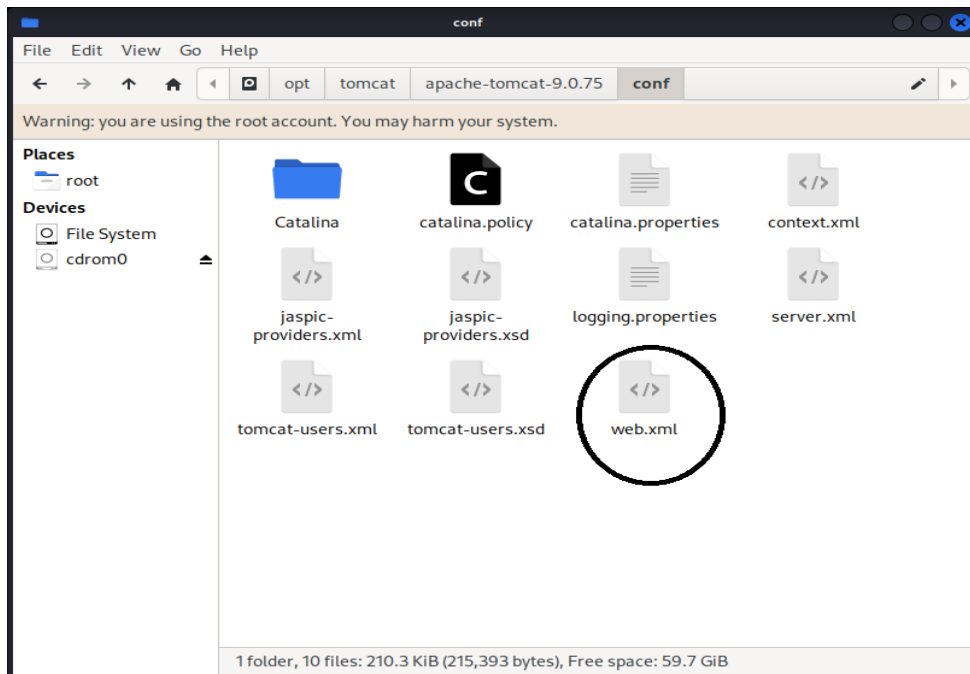


Complete all those steps, take detailed screenshots of your steps, and submit in your answer.

Subtask 2: Complete a nmap scanning with tomcat started and show that nmap finds tomcat's information. Provide screenshots of your steps including nmap scanning.

Hints: <https://nmap.org/book/vscan-examples.html>

Subtask 3: Create a custom error page to handle HTTP status 404 error and make the necessary changes in the **web.xml** file located in the following folder to prevent nmap from picking the information about the server.



The custom error page (name e.g. error.jsp) location in the ROOT folder:

/opt/tomcat/apache-tomcat-9.0.75/webapps/ROOT

Custom error page template (e.g. error.jsp):

```
<html>
<head>
<title>Page not found</title>
<link rel="icon" type="image/x-icon" href="favicon.ico">
</head>
<body>
<h1>
  <center>The Page you looking for is not available, please contact with the
  administrator.</center>
</h1>
</body>
</html>
```

Test your custom error page so that it executes instead of HTTP status 404 page of subtask-1.

Take detailed screenshots of your steps and submit with your answer including all tags, html codes, changes in web.xml file, output of your tests. Explain your answer what you have done.

Hints: you need to create a custom error page (name e.g. **error.jsp**) and modify the web.xml file in the folder **/opt/tomcat/apache-tomcat-9.0.75/conf**.

Add the following error page tags in the web.xml (at the very end) after the tag `</welcome-file-list>` and before the tag `</web-app>`:

```
<error-page>
```

```
<error-code>404</error-code >
```

```
<location>/error.jsp </location>
```

```
</error-page>
```

Repeat the tags for error-code: 403 and 500

Once you finish, Stop your tomcat server and restart using the command used before. Check status of your tomcat using **sudo systemctl status tomcat** to confirm it has started.

Test your custom error page using localhost:8080/ .

Subtask 4: Complete a **nmap scanning** and show that nmap is not revealing tomcat server's information as before, and showing information from your custom error page. Provide detailed screenshots and explanation of your work.

Please do your own research and try to collect as much information as possible. When you use information from other sources, remember to cite them, else it will be considered as plagiarism. You CAN NOT do direct copy paste even if you provide proper in-text citations and bibliographic references at the end.

Submission Requirements:

Submit one PDF file with your explanation and detailed screenshots of all subtasks as mentioned before.

Submission Due

The due for each task has been stated via its OnTrack task information dashboard.